Neuron, Volume *113*

Supplemental information

Universal statistics of hippocampal place fields across species and dimensionalities

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Supplementary materials

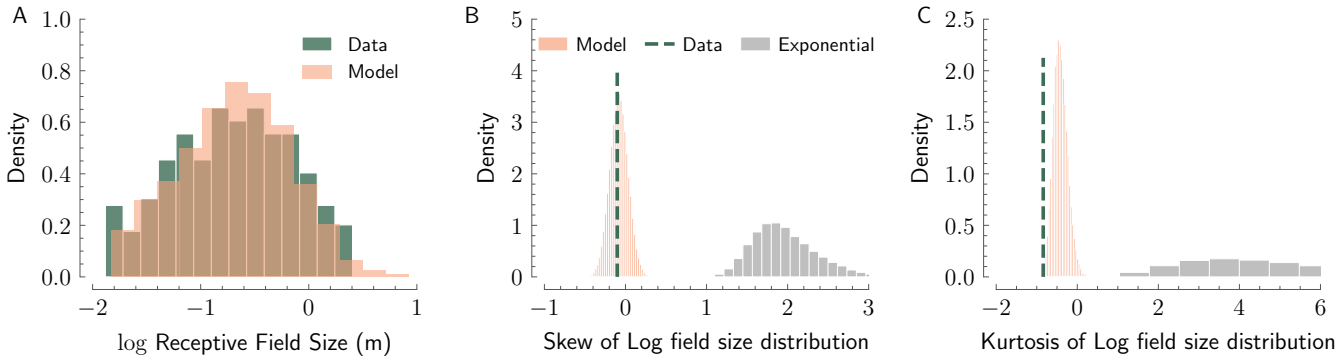


Figure S1. Field arrangements from rats in 1D environments are explained by the model. Related to Figure 1.

(A) Distribution of (log) field sizes measured in rats navigating a 48 m-long 1d maze [S1] along with distribution arising from the model ($\sigma = 0.34$, $\theta = 1.8$), fitted to match mean field size and count observed in the data. Fields in data larger than three times the standard deviation were removed as outliers. The empirical distribution matches the model better than the heuristic exponential distribution proposed in [S2] (likelihood test: $\Delta \log \text{Likelihood} = -83.1 < 0$).
(B-C) Skew/Kurtosis distribution of log field size obtained from multiple simulations of the model (orange) and the heuristic exponential distribution (gray) compared against that extracted from the experimental data (vertical green dashed line). The empirical fields were required to be larger than 15 cm in [S1]. We applied the same cutoff on the fields from to the model to match the distribution obtained from the data. Note that this affects the distribution of skew and kurtosis.

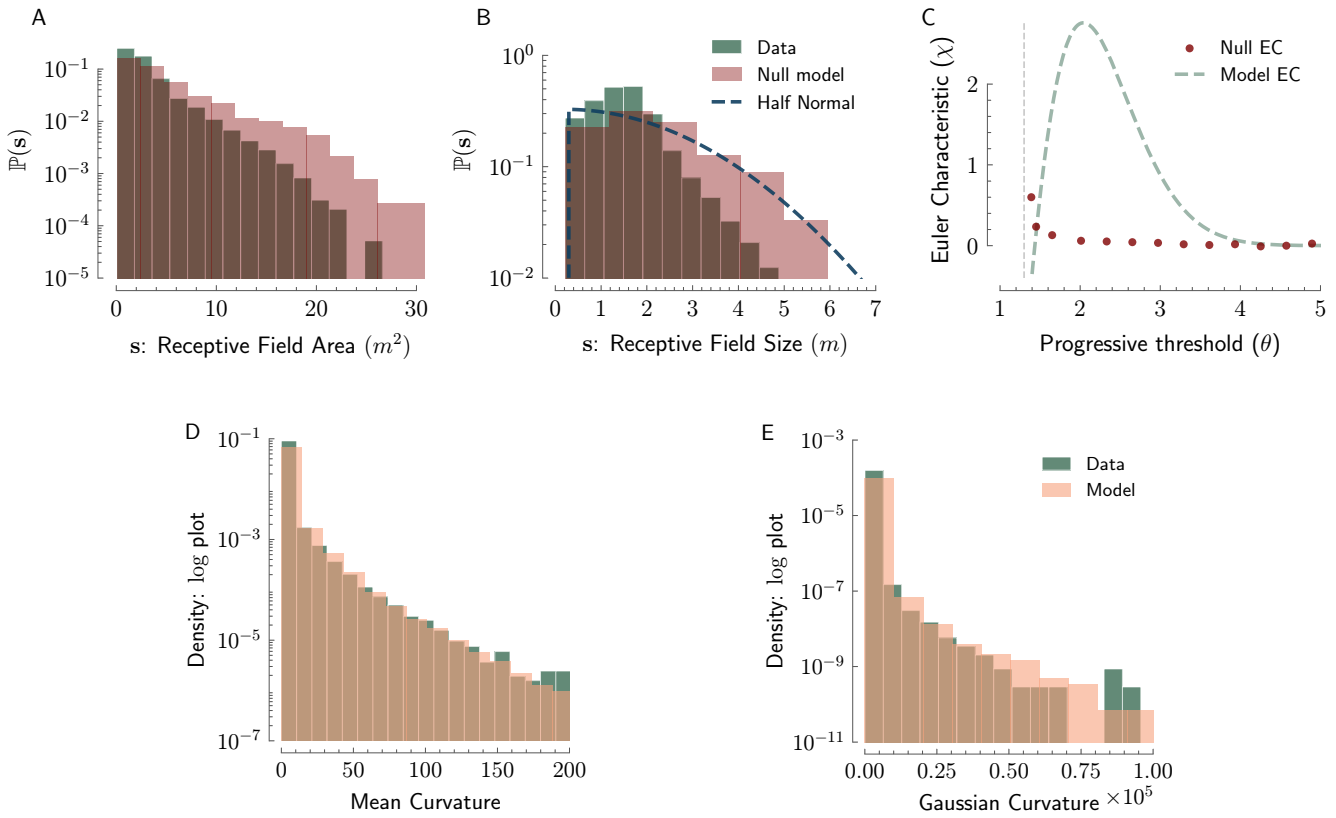


Figure S2. Further analysis of 3d place fields: comparison to alternative model with spherical fields and curvature of field iso-surfaces. Related to Figure 2 and Figure 3.

(A/B/C) Alternative model for comparison against the thresholded Gaussian process model in 3d. Firing fields in the alternative model are spherical, while the joint distribution of field volumes and maximum-firing rates exactly matches the empirical distribution. More precisely, for each field in the data, we created a corresponding field in the model with spherical support and matching volume. These fields are not allowed to overlap but are otherwise randomly placed since the measurement of field sizes in the slices and the expected Euler characteristics are independent of the field placements. The firing rate profile for each of these fields was modeled as a quadratic function with amplitude exactly matching that of the field in data.
(A) Discrepancy between areas of 2d slices through 3d fields in data and in the alternative model (logarithmic vertical scale).
(B) Discrepancy between length of 1d slices through 3d fields in data and in the alternative model (logarithmic vertical scale). In the null model, the radius of each spherical field is distributed as Rayleigh distribution, which implies that the length of active portions of 1d slices follows a half-normal distribution.
(C) Comparison between Euler characteristics of the thresholded Gaussian process model (matching the empirical data, Fig 3G), and Euler characteristics arising from the alternative model.
(D) Distribution of mean curvature and (E) Gaussian curvature of 3d place field boundaries in data and model (logarthmic vertical scale). The boundary surface for each field was first identified and represented as a two-dimensional array Z, where each point (x, y, z) of the surface was mapped to position (x, y) in the array containing the z-value of the surface points. The mean and Gaussian curvatures of the isosurface representing the boundaries of the 3D fields were then

calculated[S3] using the following operators. Gaussian curvature (K) is defined as:

$$K = \frac{\partial_{xx}\partial_{yy} - \partial_{xy}^2}{(1 + \partial_x^2 + \partial_y^2)^2}$$

and mean curvature (H) is defined as:

$$H = \frac{(1 + \partial_x^2)\partial_{yy} + (1 + \partial_y^2)\partial_{xx} - 2\partial_x\partial_y\partial_{xy}}{2(1 + \partial_x^2 + \partial_y^2)^{3/2}}$$

where ∂_x, ∂_y, ∂_{xx}, ∂_{yy}, and ∂_{xy} are the first and second-order partial derivatives of the surface represented by array Z with respect to x and y. The partial derivatives were computed using the `numpy.gradient` function, which calculates the numerical gradients of the input array.

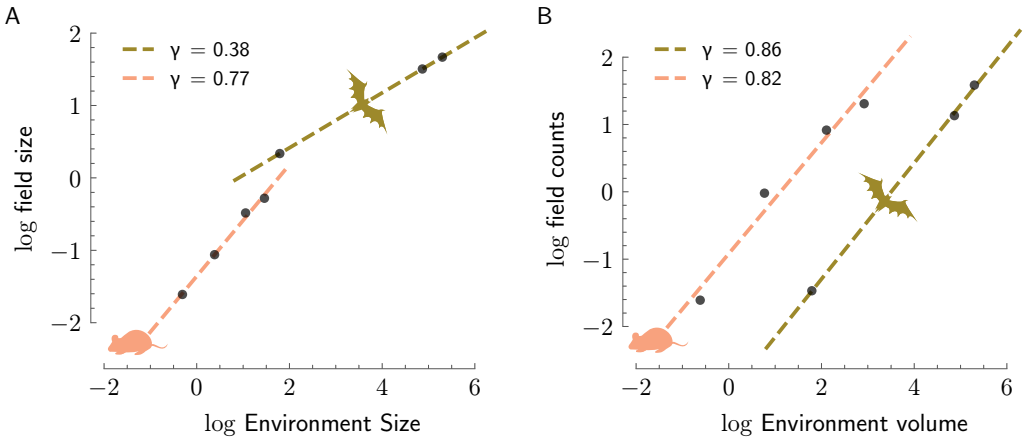


Figure S3. Trends in field statistics as a function of environment size. Related to Figure 4.

(A) Linear scaling of logarithm of linear field sizes (in m) with the logarithm of the linear environment size (in m) in 1d (bats) [S4] and 2d (rats) [S5]. In both panels, γ is the slope of the best linear fit to the data, in logarithmic scales.
(B) Linear scaling of logarithm of field counts with the logarithm of the linear environment volume (length in 1d and area in 2d) in 1d (bats) [S4] and 2d (rats) [S5]. The average propensity, defined as the average field count divided by the size of the environment (length in 1d and area in 2d) decreases weakly with environment size, exhibiting an approximate power-law dependence on the size of the environment, with an exponent $\gamma - 1$, equal approximately to -0.14 in 1d and to -0.18 in 2d.

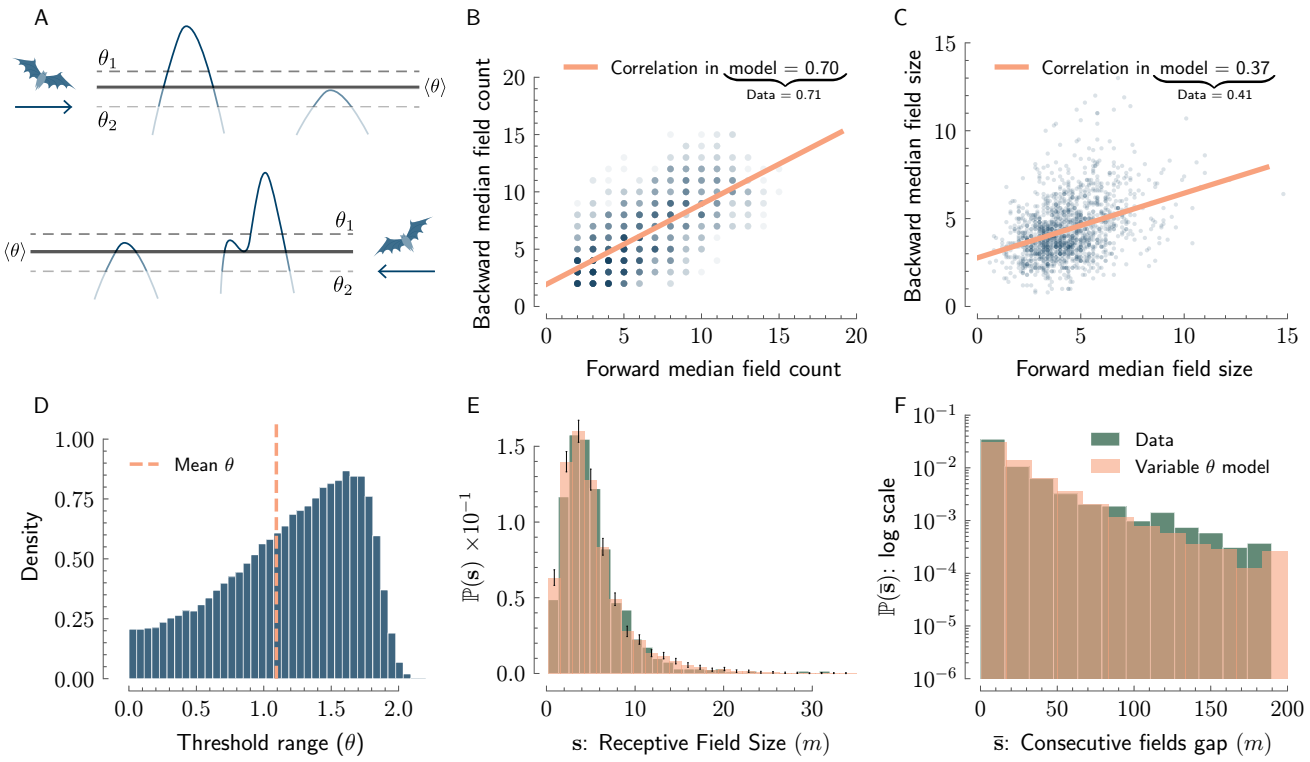
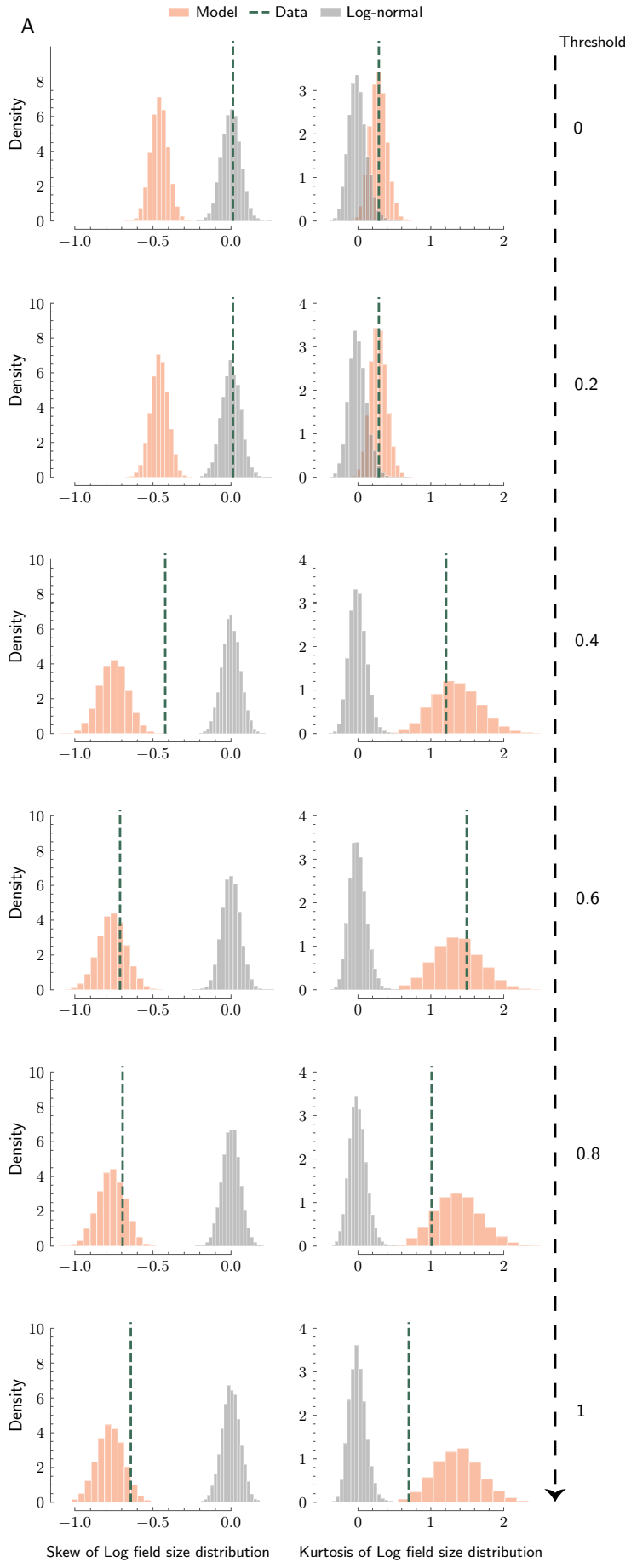


Figure S4. Threshold noise explains correlation in field statistics between flight directions of bats in a 1d environment. Related to Figure 1 and Figure 5.

(A) Illustration of how variability in the normalized threshold across cells leads to correlated field counts and field sizes in the two flight directions along the 1d tunnel. Neurons with lower threshold will have more place fields on average, when the bat flies along both directions of the tunnel. Likewise, neurons with lower threshold will have larger field size on average, when the bat flies along both directions of the tunnel.
(B) Scatter plot (blue) and correlation of field counts obtained from the model in two different samples (representing forward and backwards directions) of fields with fixed noisy threshold. With a distribution of thresholds, the model matches the empirical correlation between field counts in bats flying in the forward and backward directions (parameters chosen as explained in D).
(C) Scatter plot (blue) and correlation of median field size per cell in two different samples of fields with fixed noisy threshold. With a distribution of thresholds, the model matches the empirical correlation between field sizes in the two flying directions, while simultaneously matching the correlation in field counts (parameters chosen as explained in D).
(D) The distribution of θ in the extended model with variability in the threshold across cells. θ is sampled from a skew-normal distribution conditioned to be positive, with mean matching the single inferred θ in case of bats in 1d tunnel. The skew normal distribution has two remaining parameters after fixing the mean, which were set to match the correlation between median field count and median field size as reported in [S4].
(E) Field size distribution under the variable threshold model matches the distribution of field sizes in the data.
(F) Field gap distribution in the model matches that of the data and further explains (without additional fitting) the slightly heavier tail in the empirical distribution in comparison with the fit to an exponential distribution (compare with Fig. 1D).


A
Model
Data
Log-normal
Threshold
0
0.2
0.4
0.6
0.8
1
Density
Skew of Log field size distribution
Kurtosis of Log field size distribution

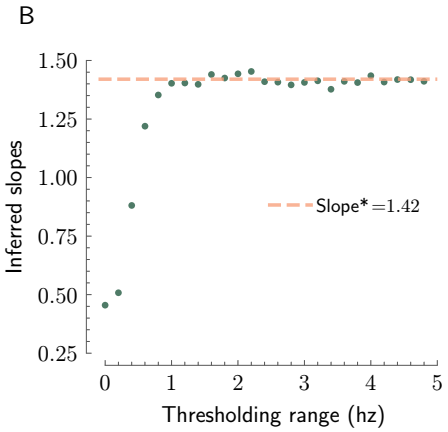

B
Inferred slopes
Slope* =1.42
Thresholding range (hz)

Figure S5. Evolution of field size distribution higher moments and inferred field size-amplitude slope across thresholds. Related to STAR Methods.

(A) The Higher moment test of skew (left column) and kurtosis (right column) between the model and log-normal is highly sensitive and changes across choices of a range of small (between $0-1$ hz) thresholds applied to the data for denoising. The choice of denoising threshold for the main figures was 0.5, and the figure shows the tests for 6 other choices.
(B) Slope of the linear fit between log field size and log maximum firing rate, evaluated for the data from bats flying in the 200 m tunnel using a range of rectification parameters. For rectification parameters above 1 Hz the slope rapidly stabilizes on a consistent value.

Methods S1
Introduction to Gaussian process. Related to Figure 1.

In this section we provide a basic introduction to Gaussian Process, and discuss the mapping between the thresholded Gaussian process and a simplified picture of neural dynamics.

Gaussian processes can be viewed as a generalization of a multivariate Gaussian variable, having a probability density function of the form

$$\phi(\mathbf{f}) = \frac{1}{(2\pi)^{L/2}|C|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{f}-\mathbf{m})^\top C^{-1}(\mathbf{f}-\mathbf{m})\right), \tag{S1}$$

where $\mathbf{f}$ is a multidimensional random variable of dimension L, $\mathbf{m}$ is its mean, C is the covariance matrix and $|C| = \det C$ is the determinant of the matrix C. Multivariate Gaussian variables have the property that for every $\alpha = (\alpha_1, \ldots, \alpha_L) \in \mathbb{R}^L$, the real valued variable $\alpha \cdot \mathbf{f} = \sum_{i=1}^{L} \alpha_i f_i$ is distributed as a univariate Gaussian variable. This property is sufficient and necessary for a multivariate variable to have the form of Eq. S10, and can therefore be used as a definition of a multivariate Gaussian distribution, instead of Eq. S10.

It is possible to generalize the concept of a Gaussian distribution over vectors to a Gaussian distribution over *functions* of a continuous variable $x \in [0, L]$: the variable x can be thought of as a continuous analogue of the discrete index that labels the elements of a random vector. More formally a random distribution over functions $f(x)$ is a Gaussian process if for every discrete set of k points, the values of the function $(f(x_1), \cdots, f(x_K))$ are jointly distributed as a multivariate Gaussian. Similarly to the discrete case, where the vector $\mathbf{m}$ and the covariance matrix C uniquely determine the distribution of a multivariate Gaussian variable, a Gaussian process is uniquely determined by its mean function

$$m(x) = \langle f(x) \rangle,$$

and its (two-variable) covariance function

$$r(x, y) = \langle [f(x) - m(x)]\,[f(y) - m(y)] \rangle\,.$$

In this work we consider stationary Gaussian processes, having the property that $r(x, y)$ is function of $x - y$ and can thus be written as $r(x - y)$. In addition, the mean $m(x)$ of a stationary Gaussian process is independent of x, and is assumed in this work to vanish. Note, specifically, that the variance of the process is constant throughout the space and is given by $r(0)$.

In our model, the underlying index $x \in [0, L]$ represents the location of the animal in its environment, and the turning curve of a neuron is obtained by applying a threshold nonlinearity to its associated instance of the Gaussian process. A simple biological interpretation of the model is that the Gaussian process represents the mean synaptic input that the neuron receives from its presynpatic partners, whose activities are themselves spatially tuned. The threshold-linear transfer function, which relates the total synaptic input to the cell's firing rate arises from the interplay of the synaptic input, the intrinsic dynamics of the neuron, and noise [S6]. Technically, in the theory of Gaussian processes, the regions in which the process exceeds a threshold are called excursion sets, and these are mapped to place fields in our model.

According to the model, at any fixed position of the animal the synaptic activation is distributed as a Gaussian variable across the neural population, and so is the synaptic activation of a single neuron across positions. This gives rise to a thresholded Gaussian distribution of firing rates. Interestingly, it was noted long ago in another brain area (IT cortex) that the distribution of firing rates is well approximated by a truncated Gaussian distribution [S7].

Methods S2
Derivation of Gaussian process input to CA1 cells. Related to Figure 1.

Here we illustrate how random summation of presynaptic inputs into CA1 cells leads to Gaussian statistics of the total synaptic input, and elucidate how the spatial correlation function of the Gaussian process is determined by the input statistics. We assume, for simplicity, completely independent weights, yet Gaussian statistics can arise also under broader conditions, in which there is partial correlation between groups of summed variables that decays sufficiently rapidly across a functional axis within the input population [S8].

Consider a population $i = 1 \dots N$ of spatially selective neurons presynpatic to a CA1 place cell with bounded spatial response function $u_i(x)$ for $x \in \mathbb{R}^D$ in D dimensions. For simplicity we consider these response functions to have zero mean, whereas in a more realistic setting u_i can stand for the deviation of the firing rate from its mean, and approximate balancing of the mean (which is interchangeable also with the choice of threshold in our model) may be obtained by other mechanisms, such as mutual lateral inhibition or homeostatic regulation of excitatory and inhibitory inputs.

We consider first, for simplicity, a scenario in which all input neurons have the same tuning curve up to translation, with preferred firing locations x_i that uniformly and densely tile the space:

$$u_i(x) = u_0(x - x_i). \tag{S2}$$

The input to the CA1 cell, $h(x)$ is a weighted linear sum of the activity of the presynpatic cell outputs

$$h(x) = \sum_{i=1}^{N} W_i u_i(x), \tag{S3}$$

where we assume that the weights W_i are drawn independently from a random distribution $p(W)$ with zero mean, and a finite second moment. Using the central limit theorem for multidimensional stimuli [S9,10], it is straightforward to see that for any discrete set of locations x_α, $h(x_i)$ are jointly distributed as multivariate normal in the large N limit. Hence, $h(x)$ is a Gaussian process in this limit.

The covariance function of the Gaussian process (which uniquely determines all of its properties) must be translationally invariant, due to our assumption of uniform tiling of inputs in the input layer. Indeed,

$$\begin{aligned}\langle h(x)h(x')\rangle &= \sum_{i,j=1}^{N} \langle W_i W_j\rangle \, u_i(x)u_i(x') = \mathrm{var}\,(w) \sum_{i=1} u_i(x)u_i(x') \\ &= \mathrm{var}\,(w) \cdot r(x' - x),\end{aligned} \tag{S4}$$

where the angular brackets represent an average over realizations of the random synaptic weights, and r is the spatial covariance of the neural population response. Here, r is proportional to the auto-correlation of the single neuron tuning curve

$$r(\,\Delta x) = \rho \int \mathrm{d}x \, u_0\,(x)\, u_0\,(x + \Delta x)\,, \tag{S5}$$

where ρ is the spatial density of receptive field centers. Specifically, if individual tuning curves are shaped as Gaussian firing fields, $r(\Delta x)$ is a Gaussian with a width equal to twice the width of the individual firing fields.

Heterogeneous tuning curves

Similar results hold for heterogeneous tuning curves. We assume that the tuning curves of individual neurons are sampled from a distribution of shapes, parameterized by a random variable α which is independently chosen for each neuron (α can be scalar or multidimensional, discrete or continuous). For each neuron i,

$$u_i(x) = u_0(x - x_i; \alpha_i), \tag{S6}$$

where x_i is drawn from a a uniform distribution in space and α_i is the shape parameter of neuron i. Under these assumptions, $h(x)$ is a sum over many independent random functions, which becomes distributed in the large N limit as a Gaussian process due to the central limit theory. A similar derivation as above yields

$$\langle h(x)h(x')\rangle = \mathrm{var}\,(w) \cdot r(x' - x), \tag{S7}$$

where $r(\Delta x)$ is proportional to the auto-correlation of the tuning curve, averaged over the distribution of receptive field shapes

$$r(\,\Delta x) = \rho \left\langle \int \mathrm{d}x\, u_0\,(x;\alpha)\, u_0\,(x + \Delta x;\alpha) \right\rangle_{\alpha}, \tag{S8}$$

where $\langle\,\rangle_\alpha$ represents an average over realizations of the shape parameter α. Qualitatively, in both scenarios discussed above, the central limit theory becomes valid when each CA1 neuron receives, within its place fields, active presynaptic inputs from many spatially selective cells.

Methods S3
Field statistics. Related to Figures 1 and 2.

An extensive body of mathematical results exist for the statistics of excursion sets of Gaussian processes. In this section we explain how a small subset of the results are obtained. Full derivations of all the results are found in Adler[S11].

Field statistics in 1D

Field counts in 1d can be exactly derived using the Kac-Rice formula [S12]:

$$\langle N\rangle = \frac{1}{2}\int_L \left\langle |f'(x)| \;\middle|\; f(x) = \theta\sqrt{r(0)} \right\rangle \mathrm{d}x, \tag{S9}$$

where $f(x)$ is a random process with a well defined derivative. Specifically, in our model $f(x)$ is a 0 mean, stationary Gaussian process, and consequently $f'(x)$ is a 0 mean stationary Gaussian process as well. To calculate the integrand, we first compute the covariance matrix associated with the joint distribution of $f(x)$ and $f'(x)$:

$$C_x = \begin{pmatrix} \langle f(x)^2\rangle & \langle f(x)f'(x)\rangle \\ \langle f'(x)f(x)\rangle & \langle f'(x)^2\rangle \end{pmatrix}. \tag{S10}$$

Using stationarity, we have:

$$(i) \quad \left\langle f(x)^2 \right\rangle = r(0)$$

$$(ii) \quad \langle f(x)f'(x)\rangle = \frac{1}{2}\frac{\partial^2}{\partial x^2}\left\langle f(x)^2\right\rangle = \frac{1}{2}\frac{\partial^2}{\partial x^2} r(0) = 0$$

$$(iii) \quad \left\langle f'(x)^2 \right\rangle = \frac{\partial^2}{\partial x_1 \partial x_2} r(x_1 - x_2)\,\big|_{x_1 = x_2} = -r''(0).$$

Thus:

$$C_x = \begin{pmatrix} r(0) & 0 \\ 0 & -r''(0) \end{pmatrix}. \tag{S11}$$

Using Eq. S20, we can calculate the expectation inside the integral

$$\begin{aligned}
&\left\langle |f'(x)| \;\middle|\; f(x) = \theta\sqrt{r(0)} \right\rangle \\
&= \frac{1}{2\pi\sqrt{-r(0)r''(0)}} \int_{-\infty}^{\infty} |f'(x)| \exp\left(-\frac{-r''(0)f(x)^2 + r(0)f'(x)^2}{-2r''(0)r(0)} \right) \delta\big(f(x) = \theta\sqrt{r(0)}\big)\, \mathrm{d}f\, \mathrm{d}f' \\
&= \frac{e^{-\theta^2/2}}{\sqrt{2\pi r(0)}} \cdot \frac{1}{\sqrt{2\pi r''(0)}} \int_{-\infty}^{\infty} |f'(x)| \exp\left(-\frac{1}{-2r''(0)} f'(x)^2 \right) \mathrm{d}f' \\
&= \frac{e^{-\theta^2/2}}{\sqrt{2\pi r(0)}} \cdot \sqrt{\frac{-2r''(0)}{\pi}} \\
&= \frac{e^{-\theta^2/2}}{\pi} \sqrt{\frac{-r''(0)}{r(0)}}.
\end{aligned}$$

Therefore, the final result for $\langle N \rangle$ is

$$\langle N \rangle = \frac{1}{2\pi} \int_L \sqrt{\frac{-r''(0)}{r(0)}} e^{-\theta^2/2}\, \mathrm{d}x = \frac{L e^{-\theta^2/2}}{2\pi\sigma}. \tag{S12}$$

We can directly relate this to the measure of propensity (number of place fields per unit length):

$$\langle \mathrm{Propensity} \rangle = \frac{e^{-\theta^2/2}}{2\pi\sigma}. \tag{S13}$$

Thus, the average propensity is suppressed by both the correlation length σ (algebraically) and the normalized threshold θ (exponentially)

Next, assuming that the environment is large compared to the correlation length, the portion of the environment in which a cell is active is $L\,(1 - \phi(\theta))$. This, together with the mean field count can be used to calculate the mean field size s and the mean gap between the fields $\bar{s}$ as

$$\langle s \rangle = 2\pi\sigma\,(1 - \phi(\theta))\, e^{\frac{\theta^2}{2}}, \tag{S14}$$

$$\langle \bar{s} \rangle = 2\pi\sigma\phi(\theta) e^{\frac{\theta^2}{2}}. \tag{S15}$$

Simulations were in excellent agreement with all these expressions.

Field statistics in higher dimensions

In higher dimensions, precise analytical expressions are not available for the mean field count and size. We relied on simulations to determine the parameters σ and θ that match the mean size and the mean field count, as described in Methods. A qualitative understanding of these relationships can be obtained using an approximation which is valid for high thresholds, obtained

by retaining the leading order term in the expression for the Euler characteristics (see [S11] for more details on this approximation):

$$\langle N \rangle = \frac{(L/\sigma)^D}{(2\pi)^{\frac{D+1}{2}}} \theta^{D-1} \exp\left(-\frac{\theta^2}{2}\right), \tag{S16}$$

where L^D is the volume of the environment.

The mean field size is then approximated by

$$\langle s \rangle = \sqrt{2\pi} \left(2\pi\sigma^2\right)^{\frac{D}{2}} \theta^{1-D} \left(1 - \phi(\theta)\right) e^{\frac{\theta^2}{2}}. \tag{S17}$$

Distributions in the high threshold regime

As described in the main text, distributions related to the field arrangements acquire a universal form in the high threshold regime. The field count is Poisson, in all dimensions, with a mean given by Eq. S23. This is equivalent to the fact that the field gaps in 1d are exponentially distributed. Similarly we can approximate the full field-size distribution as in the main text (Eq. 1) by:

$$P(s) = \frac{2\beta}{D} s^{\left(\frac{2}{D}-1\right)} \exp\left(-\beta s^{\frac{2}{D}}\right).$$

Furthermore, we can relate β to the parameters σ and θ of the model through the analytic forms for the mean field size (Eqs. S22 and S24). To that end, first, observe that $s^{2/D}$ is exponentially distributed with an expectation $\left\langle s^{2/D} \right\rangle = 1/\beta$, implying that $\langle s \rangle = \Gamma(D/2+1)\beta^{-D/2}$ using the moment generating function of the exponential distribution. Therefore:

$$\beta = \left[\Gamma\left(\frac{D}{2}+1\right) \langle s \rangle\right]^{\frac{2}{D}}. \tag{S18}$$

Methods S4
Convergence to Rayleigh distribution. Related to Figures 1, 2 and 5.

The field size distribution in our model has a stereotypical skewed form that converges to a Rayleigh distribution in the high threshold limit. To understand this behavior on very qualitative terms, we note that for high threshold crossings the fields are small compared to the correlation length of the Gaussian process, and the consecutive threshold crossings demarcating the fields are correlated (contrast this with field gaps, which in the high threshold limit are large compared to the correlation length of the process. The consecutive threshold crossings demarcating gaps are therefore *uncorrelated* in the high threshold limit, thereby leading to the observed exponential distribution).

A full mathematical characterization of the threshold crossing's joint distribution – when the crossings are nearby as in the case of the place fields, is still an open problem for general Gaussian processes, yet some results are available. Under broad regularity conditions, it has been mathematically established that, when the threshold crossings are nearby, the threshold crossings tend to repel each other [S13]. Moreover, it is established that the probability of small distance δx between the consecutive nearby threshold crossing $p(\delta x)$ behaves, for small δx as

$$p(\delta x) \propto \delta x. \tag{S19}$$

Effectively this implies that the probability of small field size approaches 0 when $\delta x \to 0$. The repulsion between consecutive crossings can be understood intuitively from the observation that

for there to be another threshold crossing after a threshold crossing, the derivative of the process must change sign. Since we assume a smooth process, the sign change of the derivative will only occur after some distance from the first threshold crossing, effectively suppressing the probability of having very small excursion sets.

It is also straightforward to evaluate how the probability density function for the gap size x decays to zero for large x. Suppose that a stationary Gaussian process $f(x)$ with unit variance crossed the threshold θ at the origin i.e, $f(0) = \theta$. We will calculate the probability that the process is still above the threshold at x, i.e, $P\{f(x) > \theta \mid f(0) = \theta\}$. To that end, we start by noting that $f(0)$ and $f(x)$ are jointly Gaussian with covariance $r(x)$, and by Gaussian conditioning we get:

$$(f(x) \mid f(0) = \theta) \sim \mathcal{N}\left(\theta r(x), 1 - r(x)^2\right). \tag{S20}$$

We can then express the probability of $f(x)$ being above threshold by

$$\begin{aligned} P\{f(x) > \theta \mid f(0) = \theta\} &= 1 - \Phi\left(\frac{\theta - \theta r(x)}{\sqrt{1 - r(x)^2}}\right) \\ &\leq \mathcal{O}\left(\exp\left[-\frac{1}{2}\frac{\theta^2\left(1 - r(x)\right)^2}{1 - r(x)^2}\right]\right) \\ &= \mathcal{O}\left(\exp\left[-\frac{\theta^2}{2}\frac{1 - r(x)}{1 + r(x)}\right]\right) \\ &\approx \mathcal{O}\left(\exp\left[-\frac{\theta^2}{4}\left(\frac{x}{\sigma}\right)^2\right]\right) \\ \implies P\{f(x) > \theta \mid f(0) = \theta\} &\leq \mathcal{O}\left(\exp\left[-\left(\frac{\theta x}{2\sigma}\right)^2\right]\right), \end{aligned} \tag{S21}$$

where we have used the Taylor expansion of the correlation function $r(x) \approx 1 - \frac{1}{2}\left(\frac{x}{\sigma}\right)^2$ which holds in the high threshold regime. We can conclude from this derivation that the tail of the field size distribution decays as a function of x at least as fast as $\exp\left[-\left(\frac{\theta x}{2\sigma}\right)^2\right]$.

Together, these results on the dependence of the probability density function on x – for small x (linear dependence on x) and for large x (decay as a squared exponential function of x) explain the key features of the Rayleigh distribution. Next, we will sketch a more formal derivation of the field size distribution in the high threshold limit.

We start with a theorem concerning the behavior of a Guassian process near a maximum that occurs above a high threshold [S11]. Consider a centered stationary Gaussian process $f(x)$ with unit variance. Assume that $f(x)$ has a local maximum at $x = 0$ that exceeds a threshold θ by a height h, such that $f(0) = \theta + h$. When θ approaches infinity and for small values of $|x|$, the process can be approximated as

$$f(x) \approx f(0) + f(0)\frac{r''(0)}{2}x^2. \tag{S22}$$

This theorem simply states that at the limit of a high threshold crossing, in the neighborhood of the maxima of the firing rate, the fields are well approximated by a parabola with curvature that concentrates at $r''(0)$. This allows us to think of the field as parabolic and express the relationship between the size of the excursion s and the field height h:

$$\lim_{\theta \to \infty} s = 2\sigma\sqrt{\frac{h}{\theta}}. \tag{S23}$$

(Note that this results implies a power law relating the field size and amplitude with exponent 2, in agreement with the trends seen in Fig. 5. Note also that field shapes can be well approximated by a parabola for $\theta = 2.4$.)

Equipped with Eq. (S32), if we now find the distribution for the height of the fields h, we can then approximate the distribution of the field sizes. To that end, we define the height variable $H_\theta = f(0) - \theta$ and observe that

$$P\{H_\theta \geq h \mid O_\theta \geq 0\} = \frac{\langle M(\theta + h)\rangle}{\langle M(\theta)\rangle}, \tag{S24}$$

where $\langle M(\theta)\rangle$ represents mean number of maxima of a Gaussian process above the threshold θ. Intuitively, the expression relates the tail of the height distribution to the ratio between the mean number of maxima above $\theta + h$ and the mean number of maxima above θ.

In the high threshold regime, each field has a single maximum, and we can replace the mean number of maxima $M(\theta)$ with the field count $N(\theta)$, which we know from the Kac-Rice formula, Eq. (S21). We thus obtain:

$$P\{H_\theta \geq h \mid O_\theta \geq 0\} = \frac{\exp\left(-(\theta + h)^2/2\right)}{\exp\left(-\theta^2/2\right)},$$

and in the high threshold limit,

$$\begin{aligned} P\{H_\theta \geq h \mid O_\theta \geq 0\} &= \frac{\exp\left(-(\theta + h)^2/2\right)}{\exp(-\theta^2/2)} \\ &= \frac{\exp\left(-(\theta^2 + 2h\theta + h^2)/2\right)}{\exp(-\theta^2/2)} \\ &= \frac{\exp(-\theta^2/2) \cdot \exp(-(2h + h^2/\theta^2)/2)}{\exp(-\theta^2/2)} \\ &= \exp(-h\theta + h^2/2). \end{aligned}$$

Viewed as a function of $\hat{h} = \theta h$, the distribution approaches a non-trivial stationary limit when $\theta \to \infty$,

$$\lim_{\theta\to\infty} P\{H_\theta \geq \hat{h}/\theta \mid O_\theta \geq 0\} = \lim_{\theta\to\infty} \exp(-\hat{h} + \hat{h}^2/2\theta^2) = \exp(-\hat{h}). \tag{S25}$$

We conclude that the height is distributed as an exponential random variable with mean $1/\theta$, and since the field size is related to the height, we can show using a change of variables to show that the field size is distributed as a Rayleigh distribution. First recall that

$$\lim_{\theta\to\infty} s = 2\sigma\sqrt{\frac{h}{\theta}} = \frac{2\sigma}{\theta}\sqrt{\hat{h}} \implies \hat{h} = \frac{\theta^2 s^2}{4\sigma^2},$$

where $\sqrt{\hat{h}} \sim \exp(1)$. Then:

$$\begin{aligned} P_{\hat{h}}(\hat{h}) = e^{-\hat{h}} \quad ; \quad \frac{d\hat{h}}{ds} &= \frac{\theta^2 s}{2\sigma^2} \\ \implies P_s(s) = P_{\hat{h}}(\hat{h}(s)) \left|\frac{d\hat{h}}{ds}\right| &= \exp\left(-\frac{\theta^2 s^2}{4\sigma^2}\right) \cdot \frac{\theta^2 s}{2\sigma^2}, \end{aligned}$$

which is a Rayleigh distribution.

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